DBMS IMPORTANT QUESTIONS

UNIT-WISE LIST

Unit-wise list based on analysis of the provided previous-year question papers

★ = Most important / high-priority question

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UNIT 1: Database System Concepts and Architecture

1. ★ Explain the components of DBMS.
2. ★ Explain the three-level architecture of DBMS with a diagram.
3. ★ Explain the three views of data: External, Conceptual and Internal.
4. Explain client-server architecture of DBMS.
5. ★ Define database schema. Explain logical and physical database schemas.
6. ★ Explain the advantages of DBMS over the file processing system.

UNIT 2: Data Models and Keys

1. ★ Define a data model. Explain different types of data models with examples.
2. ★ Draw an E-R diagram for a University Management System.
3. ★ Draw an E-R diagram for a Library Management System.
4. ★ Differentiate between Primary Key and Foreign Key.
5. ★ Differentiate between Strong Entity Set and Weak Entity Set.
6. Explain Primary, Candidate, Super and Foreign Keys with examples.

UNIT 3: Relational Model

1. Explain the Relational Data Model with an example.
2. ★ Explain Union, Intersection and Difference operations in relational algebra with examples.
3. ★ Explain Selection and Projection operations with examples.
4. ★ Explain Cartesian Product, Join and Division operations in relational algebra.
5. Discuss the conversion of an E-R model into a Relational Model.
6. ★ Explain Relational Database Constraints with examples.

UNIT 4: Relational Database Design

1. ★ Explain normalization with a suitable example.
2. ★ Explain First Normal Form (1NF) and Second Normal Form (2NF).
3. ★ Explain Third Normal Form (3NF) with an example.
4. ★ Explain Functional Dependency with an example.

5. ★ Explain Boyce-Codd Normal Form (BCNF) with an example.
6. Differentiate between 3NF and BCNF.

UNIT 5: SQL / MySQL

1. ★ Explain different data types used in SQL with examples.
2. ★ Explain DDL, DML and DQL commands with examples.
3. ★ Explain the DISTINCT clause with an example.
4. ★ Explain the ORDER BY clause with an example.
5. ★ Explain Arithmetic, Comparison and Logical Operators in SQL with examples.
6. Explain SQL commands used for data manipulation with examples.

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